THE BIPLANAR CLIQUE BLOW-UPS OF THE FIVE-CYCLE

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Abstract. We prove that a clique blow-up of the five-cycle is biplanar if and only if it has at most sixteen vertices and clique number at most eight. Consequently every biplanar clique blow-up of the five-cycle is 8-colourable, which answers a question of Trivedi. The non-biplanar cases are refuted by a propositional encoding of biplanarity, strengthened by a contraction argument for K_9, whose unsatisfiability certificates are checked in Lean 4.

1. Introduction

A graph is *biplanar* if its edge set is the union of the edge sets of two planar graphs on the same vertex set. Ringel's Earth–Moon problem [13] asks for the largest chromatic number of a biplanar graph. A biplanar graph on $n \geq 3$ vertices has at most $6n - 12$ edges, so biplanar graphs are 11-degenerate and 12 colours always suffice; Sulanke's graph, the join of a five-cycle with K_6, is biplanar and 9-chromatic [7, 10]. The answer is known only to lie in $\{9, 10, 11, 12\}$ [8].

Graphs of small independence number are the natural candidates for large chromatic number, and clique blow-ups of odd cycles are the simplest of them. For $w = (w_1, \dots, w_5) \in \mathbb{Z}^5_{\geq 1}$ let $C_5[w]$ be the graph whose vertex set is the disjoint union of *bags* $A_1, \dots, A_5$ with $|A_i| = w_i$, two distinct vertices being adjacent when their bags are equal or consecutive on the five-cycle. Indices of bags are read modulo 5 throughout, and we write

$$n(w) = w_1 + \cdots + w_5, \qquad s(w) = \max_{1 \leq i \leq 5} (w_i + w_{i+1}).$$

Thus $n(w)$ is the number of vertices of $C_5[w]$ and $s(w)$ its clique number. Every independent set of $C_5[w]$ meets at most two bags, so $\chi(C_5[w]) \geq \lceil n(w)/2 \rceil$. Boutin, Gethner and Sulanke proved that the blow-ups $C_m[K_3]$ are biplanar for every $m \geq 4$ [4], and Albertson, Boutin and Gethner asked about thickness and chromatic number of unbalanced blow-ups [1]. Trivedi [16] showed that

$$\chi(C_5[w]) = \max\{s(w), \lceil n(w)/2 \rceil\}, \tag{1.1}$$

deduced that every biplanar $C_5[w]$ is 9-colourable, and asked whether the value 9 is attained [16, Question 7.3]. By (1.1) this happens only for $n(w) \in \{17, 18\}$, and he reduced the question to 35 weight vectors. Kirchweger, Scheucher and Szeider had earlier shown with a long SAT computation that $C_5[4, 4, 4, 4, 3]$ is not biplanar [11], and Szeider has since proved that every biplanar graph of independence number at most two has at most eighteen vertices and is 9-colourable [15]. Since 9-chromatic biplanar graphs of independence number two exist on seventeen vertices [9], neither result decides Trivedi's question.

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Theorem 1.1. *Let $w \in \mathbb{Z}^5_{\geq 1}$. The graph $C_5[w]$ is biplanar if and only if $s(w) \leq 8$ and $n(w) \leq 16$.*

Corollary 1.2. *Every biplanar clique blow-up of the five-cycle is 8-colourable, and the bound is attained by the biplanar graph $C_5[3,3,3,3,4]$. In particular no biplanar clique blow-up of the five-cycle is 9-chromatic, and Question 7.3 of [16] has a negative answer.*

Proof. If $C_5[w]$ is biplanar then $s(w) \leq 8$ and $n(w) \leq 16$ by Theorem 1.1, so $\chi(C_5[w]) \leq 8$ by (1.1). The graph $C_5[3,3,3,3,4]$ satisfies both conditions and has sixteen vertices, so it is biplanar and $\chi \geq \lceil 16/2 \rceil = 8$. □

The condition $s(w) \leq 8$ is necessary because two consecutive bags span a clique of order $w_i + w_{i+1}$, and K_9 is not biplanar [3, 18]. Section 2 reduces the rest of Theorem 1.1 to 36 graphs on sixteen vertices, which we draw, and 26 graphs on seventeen vertices, which we refute. A biplanar graph on seventeen vertices has at most 90 edges while ours have between 71 and 80, so counting gives nothing, and a direct search over pairs of planar graphs is hopeless. The refutation rests on Lemma 3.1: if a component of one layer outside an 8-clique Q sees enough of Q, contracting it produces a ninth vertex adjacent to all of Q, and the two layers then contain K_9. Section 4 builds this into a propositional formula, and Section 5 describes the unsatisfiability certificates and how they are checked.[1]

2. Reduction to finitely many graphs

The dihedral group of the five-cycle acts on $\mathbb{Z}^5_{\geq 1}$ by permuting coordinates, and $C_5[w]$ depends on w only up to this action. If $v \leq w$ coordinatewise then $C_5[v]$ is an induced subgraph of $C_5[w]$, and subgraphs of biplanar graphs are biplanar.

Lemma 2.1. *Let $w \in \mathbb{Z}^5_{\geq 1}$ with $s(w) \leq 8$.*

(a) *If $n(w) \geq 17$, there is $v \in \mathbb{Z}^5_{\geq 1}$ with $v \leq w$, $n(v) = 17$ and $s(v) \leq 8$.*
(b) *If $n(w) \leq 16$, there is $v \in \mathbb{Z}^5_{\geq 1}$ with $v \geq w$, $n(v) = 16$ and $s(v) \leq 8$.*

Proof. (a) While $n(w) > 17$ some coordinate is at least 2; lowering it by one keeps all coordinates positive and does not increase s.

(b) Call the pair $\{i, i+1\}$ *tight* if $w_i + w_{i+1} = 8$. If no coordinate can be raised by one without breaking $s(w) \leq 8$, every i lies in a tight pair. The five-cycle has no three pairwise intersecting edges, so a set of edges covering all five vertices contains two disjoint ones, say $\{i, i+1\}$ and $\{i+2, i+3\}$; then $n(w) = 8 + 8 + w_{i+4} \geq 17$. Hence when $n(w) \leq 16$ some coordinate can be raised, and we repeat until $n = 16$. □

Listing the vectors directly gives the following.

Lemma 2.2. *Up to the dihedral action there are exactly 26 vectors $w \in \mathbb{Z}^5_{\geq 1}$ with $n(w) = 17$ and $s(w) \leq 8$, and exactly 36 with $n(w) = 16$ and $s(w) \leq 8$. They are listed in Tables 2 and 1.*

Since $s(w) \leq 8$ and all coordinates are positive, every coordinate is at most 7, so the lists come from a finite enumeration. The 26 vectors of the first list all satisfy Trivedi's further conditions $\sum_i w_i w_{i+1} \leq 4n - 8$ and $e \leq 6n - 12$, and they are exactly his seventeen-vertex candidates; his nine eighteen-vertex candidates each contain one of them.

Proposition 2.3. *For each of the 36 vectors of Table 1, the graph $C_5[w]$ is biplanar.*

[1] The drawings, the colourings, the reduction to finitely many weight vectors and all 26 unsatisfiability certificates are checked in Lean 4; Section 5 lists what is checked and what is assumed.

$1,1,6,1,7$	$1,1,6,2,6$	$1,2,5,1,7$	$1,2,5,2,6$	$1,2,5,3,5$	$1,2,6,1,6$
$1,2,6,2,5$	$1,3,4,1,7$	$1,3,4,2,6$	$1,3,4,3,5$	$1,3,4,4,4$	$1,3,5,1,6$
$1,3,5,2,5$	$1,3,5,3,4$	$1,4,3,2,6$	$1,4,3,3,5$	$1,4,3,4,4$	$1,4,4,1,6$
$1,4,4,2,5$	$1,5,2,2,6$	$1,5,2,3,5$	$2,2,4,2,6$	$2,2,4,3,5$	$2,2,4,4,4$
$2,2,5,2,5$	$2,3,3,2,6$	$2,3,3,3,5$	$2,3,3,4,4$	$2,3,3,5,3$	$2,3,4,2,5$
$2,3,4,3,4$	$2,3,4,4,3$	$2,3,5,2,4$	$2,4,2,4,4$	$2,4,3,3,4$	$3,3,3,3,4$

TABLE 1. The 36 weight vectors with $n(w) = 16$ and $s(w) \le 8$, up to the dihedral action. For each of them $C_5[w]$ is the union of two planar graphs with explicit straight-line drawings.

Proof. For each vector we give two edge sets L_1, L_2 with $L_1 \cup L_2 = E(C_5[w])$ and, for each L_j, an injective placement of the sixteen vertices at lattice points of $[0,28] \times [0,14]$ such that no vertex lies on a segment between two others joined in L_j and no two segments of L_j without a common endpoint meet. Each L_j is then a plane straight-line drawing. The data are part of the archive [2]; every condition is a sign test on a 2×2 integer determinant. Figure 1 shows a redrawing of the two layers for $w = (3,3,3,3,4)$. □

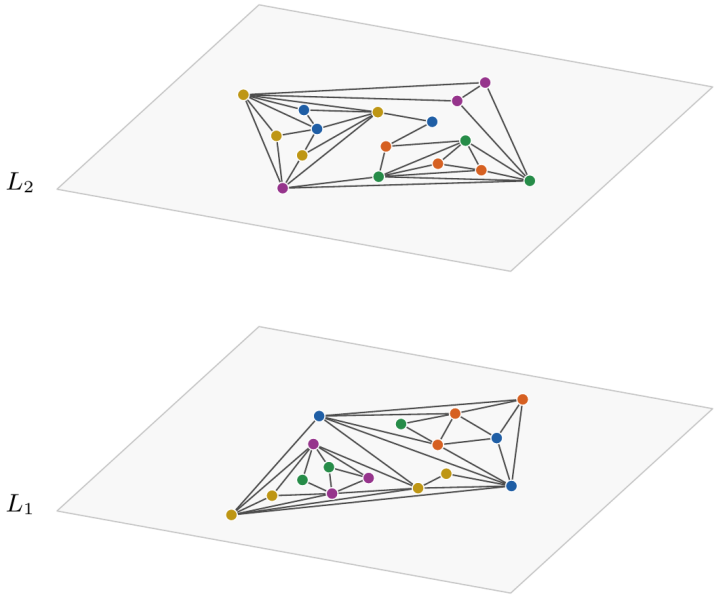


FIGURE 1. The two planar layers L_1 and L_2 of $C_5[3,3,3,3,4]$. Vertices are coloured by bag; every edge of the graph lies in one of the two layers.

3. A contraction lemma for K_9

From now on G is a graph whose edge set is $E(L_1) \cup E(L_2)$ for two planar graphs L_1, L_2 on $V(G)$, its *layers*. For $S \subseteq V(G)$ we write $L_j[S]$ for the induced subgraph, and for a set C of vertices $N_j(C)$ is the set of vertices outside C with an L_j-neighbour in C.

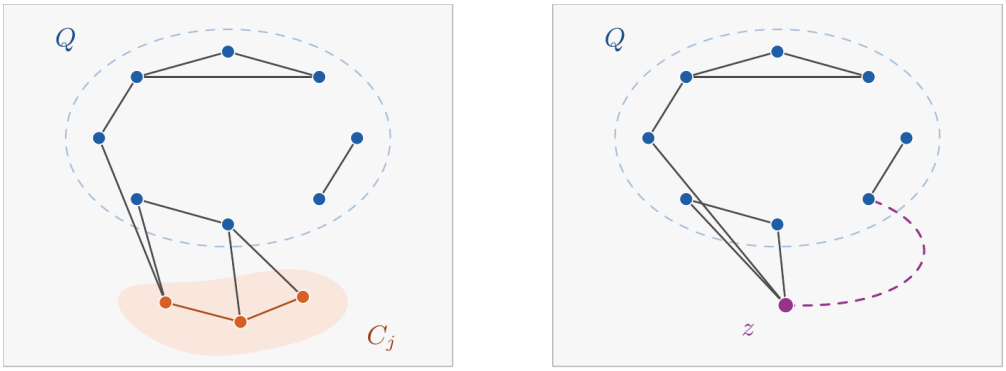
Lemma 3.1. *Let Q be a clique of G with $|Q| = 8$ and put $W = V(G) \setminus Q$. For $j = 1,2$ let C_j be either empty or the vertex set of a component of $L_j[W]$, and let $F_j \subseteq Q$ contain at most*

one vertex from each component of $L_j[Q]$ that is disjoint from $N_j(C_j)$. Then

$$(N_1(C_1) \cup F_1) \cup (N_2(C_2) \cup F_2) \neq Q.$$

Proof. Suppose the union is Q. Fix j. If $C_j \neq \emptyset$, contract the connected set C_j of $L_j[Q \cup C_j]$ to a single vertex z; if $C_j = \emptyset$, add an isolated vertex z to $L_j[Q]$. The result H_j is planar, as a minor of L_j, and z is adjacent in H_j exactly to $N_j(C_j) \cap Q = N_j(C_j)$.[2] A component K of $L_j[Q]$ disjoint from $N_j(C_j)$ has no edge to z, so it is a component of H_j not containing z. Joining z to one vertex of such a component keeps the graph planar: draw K inside a face incident with z, with the chosen vertex on the outer face of K. Doing this for the components that meet F_j, one at a time, gives a planar graph H'_j on $Q \cup \{z\}$ in which z is adjacent to $N_j(C_j) \cup F_j$; a component already joined to z is never used twice, since F_j has at most one vertex in it. Figure 2 shows the construction.

Every edge of the clique Q lies in L_1 or in L_2, hence in $L_1[Q] \subseteq H'_1$ or in $L_2[Q] \subseteq H'_2$, and z is adjacent to every vertex of Q in H'_1 or in H'_2 by assumption. So $H'_1 \cup H'_2$ contains a K_9 on $Q \cup \{z\}$ and is the union of two planar graphs, contradicting the theorem of Battle, Harary and Kodama and of Tutte [3, 18]. □



(a) (b)

FIGURE 2. Lemma 3.1 in one layer. (a) A component C_j of $L_j[W]$ and its neighbours in the clique Q. (b) After contraction z is adjacent to $N_j(C_j)$; the dashed edge joins z to a component of $L_j[Q]$ that C_j does not reach, and keeps the layer planar.

In $C_5[w]$ with $s(w) \leq 8$, each pair of consecutive bags with $w_i + w_{i+1} = 8$ is such a clique Q, and W is the union of the other three bags.

4. The encoding

We use Schnyder's characterisation of planarity in the following form.

Theorem 4.1 (Schnyder [14])**.** *If a graph H is planar, there are three linear orders $<_1, <_2, <_3$ on $V(H)$ such that for every edge uv and every vertex $x \notin \{u, v\}$ there is an i with $u <_i x$ and $v <_i x$.*

[2]Every neighbour of a component of $L_j[W]$ lies in Q, since another neighbour in W would belong to the component.

Schnyder proved that the incidence poset of vertices and edges of a planar graph has dimension at most three. Restricting a realizer to the vertices gives the orders above: an edge $e = uv$ and a vertex $x \notin e$ are incomparable, so some linear extension puts x above e and hence above u and v.[3] The same condition is the planarity encoding used in [11].

Fix w with $n = n(w)$, number the vertices $0, \dots, n-1$ bag by bag, and let E be the edge set of $C_5[w]$. The formula $\Phi(w)$ has the following variables: $o^{j,i}_{ab}$ for $j \in \{1,2\}$, $i \in \{1,2,3\}$ and $a < b$, meaning $a <_{j,i} b$ in the ith order of layer j (with $o^{j,i}_{ba}$ standing for $\neg o^{j,i}_{ab}$); ℓ^j_e for $e \in E$, meaning $e \in L_j$; and auxiliary variables, each defined by a conjunction or disjunction of earlier literals through the usual three-clause definitions [17]. Its clauses are:

(E1) transitivity of each order $<_{j,i}$;
(E2) $\ell^1_e \vee \ell^2_e$ for every $e \in E$;
(E3) for every $e = uv \in E$, every j and every $x \notin \{u, v\}$: $\ell^j_e \Rightarrow \bigvee_i (o^{j,i}_{ux} \wedge o^{j,i}_{vx})$;
(E4) for any two vertices $a < b$ that are consecutive in the numbering of the same bag, $a <_{1,1} b$;
(E5) the clauses of Lemma 3.1 described below, for every pair of consecutive bags $Q = A_i \cup A_{i+1}$ with $|Q| = 8$.

For (E5) fix such a Q and $W = V \setminus Q$. Connectivity inside a set S is expressed by the variables

$$r^{j,S,0}_{xy} = [x = y], \qquad r^{j,S,t+1}_{xy} = r^{j,S,t}_{xy} \vee \bigvee_{z \in S,\ zy \in E} (r^{j,S,t}_{xz} \wedge \ell^j_{zy}),$$

and $R^{j,S}_{xy} = r^{j,S,|S|-1}_{xy}$ holds exactly when x and y lie in one component of $L_j[S]$. For $x \in W \cup \{*\}$ and $q \in Q$ put

$$\mathsf{att}^j_{x,q} = \bigvee_{y \in W,\ yq \in E} (R^{j,W}_{xy} \wedge \ell^j_{yq}), \qquad \mathsf{att}^j_{*,q} = \bot, \qquad \mathsf{free}^j_{x,u} = \bigwedge_{q \in Q} \neg(R^{j,Q}_{uq} \wedge \mathsf{att}^j_{x,q}).$$

So $\mathsf{att}^j_{x,q}$ says that $q \in N_j(C)$ for the component C of x in $L_j[W]$ (with $C = \emptyset$ for $x = *$), and $\mathsf{free}^j_{x,u}$ that the component of u in $L_j[Q]$ is disjoint from $N_j(C)$. For every pair $x_1, x_2 \in W \cup \{*\}$ let $c_q = \mathsf{att}^1_{x_1,q} \vee \mathsf{att}^2_{x_2,q}$, and for $T \subseteq Q$ write $c_{Q \setminus T}$ for $\bigwedge_{q \in Q \setminus T} c_q$. The clauses (E5) forbid

$$c_Q; \qquad c_{Q \setminus \{u\}} \wedge \mathsf{free}^j_{x_j,u} \quad (u \in Q,\ j = 1,2); \qquad c_{Q \setminus \{u,u'\}} \wedge \mathsf{free}^1_{x_1,u} \wedge \mathsf{free}^2_{x_2,u'} \quad (u \neq u');$$
$$c_{Q \setminus \{u,u'\}} \wedge \mathsf{free}^j_{x_j,u} \wedge \mathsf{free}^j_{x_j,u'} \wedge \neg R^{j,Q}_{uu'} \quad (u \neq u',\ j = 1,2).$$

Proposition 4.2. *If $C_5[w]$ is biplanar, then $\Phi(w)$ is satisfiable.*

Proof. Let L_1, L_2 be planar layers of $C_5[w]$. Every permutation of a bag is an automorphism of $C_5[w]$, so after renaming the vertices inside each bag we may assume that the orders of Theorem 4.1 for L_1 list each bag in increasing order in $<_{1,1}$; renaming carries layers to layers and orders to orders. Set ℓ^j_e true exactly when $e \in E(L_j)$, the variables o from the six orders, and every auxiliary variable to the value of its definition. Then (E1), (E2) and (E4) hold, and (E3) holds by Theorem 4.1.

By induction on t, $r^{j,S,t}_{xy}$ is true exactly when $L_j[S]$ has an x–y walk of length at most t, so $R^{j,S}_{xy}$ is reachability in $L_j[S]$, and att and free have the meaning stated above. Given x_1, x_2, let C_j be the component of x_j in $L_j[W]$, or $\emptyset$ if $x_j = *$. Then c_q is true exactly when $q \in N_1(C_1) \cup N_2(C_2)$. A violated clause of (E5) would give sets F_1, F_2 with at most two elements in total, each element in a component of $L_j[Q]$ disjoint from $N_j(C_j)$ and two elements of F_j in different components, such that $N_1(C_1) \cup F_1 \cup N_2(C_2) \cup F_2 = Q$. Lemma 3.1 rules this out. □

[3] The converse also holds, but it is not used: the biplanar graphs of Proposition 2.3 are certified by drawings.

The formula carries one clause family for each 8-clique and each pair x_1, x_2, and only covers with at most two added edges; that is all the computation needed. The variables $o_{ab}^{j,i}$ are indexed arithmetically, so some indices do not occur.

5. Certificates

Theorem 5.1. *For each of the* 26 *vectors w of Table 2, $\Phi(w)$ is unsatisfiable. Hence none of the graphs $C_5[w]$ is biplanar.*

The formulas are produced by a Lean 4 function from w, and the clause counts in Table 2 are those of its output. CaDiCaL 3.0.1 [5] refutes each formula on one core and writes a proof in the LRAT format [6], a clausal proof in which every derived clause carries the hints that justify it. The proofs were shortened with lrat-trim, which keeps only the clauses used in the final refutation, and each trimmed proof was then checked against the formula by the LRAT checker of the Lean 4 standard library [12], which is proved sound in Lean. The result of each check is a Lean theorem stating that the formula is unsatisfiable.

Table 2. The 26 weight vectors with $n(w) = 17$ and $s(w) \le 8$, up to the dihedral action, with the size of $\Phi(w)$, the solving time of CaDiCaL, the size of the trimmed LRAT proof and the time of the Lean check.

The argument rests on the following, which are not formalised: Theorem 4.1; the facts that minors of planar graphs are planar and that joining two components of a planar graph by an edge keeps it planar; the non-biplanarity of K_9 [3, 18]; and Proposition 4.2, whose proof is above. The Lean checks of the certificates evaluate the verified checker by compiled code, so they also trust the Lean compiler. Proposition 2.3, Lemma 2.2, the enumeration behind Lemma 2.1 and 8-colourings of the 36 graphs of Table 1 are checked in Lean by kernel evaluation.[4]

6. Proof of Theorem 1.1

Let $w \in \mathbb{Z}^5_{\ge 1}$. If $s(w) \ge 9$ then $C_5[w]$ contains K_9 and is not biplanar. Suppose $s(w) \le 8$. If $n(w) \ge 17$, Lemma 2.1(a) gives $v \le w$ with $n(v) = 17$ and $s(v) \le 8$; up to symmetry v is in Table 2, so $C_5[v]$ is not biplanar by Theorem 5.1, and neither is its supergraph $C_5[w]$. If $n(w) \le 16$, Lemma 2.1(b) gives $v \ge w$ in Table 1 up to symmetry, and $C_5[w]$ is a subgraph of the biplanar graph $C_5[v]$ of Proposition 2.3. □

Remark 6.1. The graph $C_5[4,4,4,4,3]$ of [11] contains $C_5[3,3,3,4,4]$, one of the graphs of Table 2. Without the clauses (E5) the formulas of Table 2 were not refuted in hours of solving, while with them the slowest takes

Data availability. The weight vectors, drawings, colourings, formulas and certificates, together with the Lean 4 development and independent checks, are archived with the source of this paper [2].

Use of generative AI. During the preparation of this work the author used Claude (Anthropic) to search the literature, to run and check the computations, and to draft the text. The author checked every step of the proofs and every reference, and takes full responsibility for the content.

[4]The colourings give a second proof of the bound $\chi \le 8$ in Corollary 1.2 that does not use (1.1): a biplanar $C_5[w]$ is a subgraph of one of the 36 graphs.

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